

ESG DASHBOARD Using AI

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The Trade-offs of AI-Powered Dashboard Development

Pros

- **Time Efficient:** Significantly reduces the time required to build dashboards. For example, a dashboard containing 10 charts can often be completed within a few hours.
- **No Coding Required:** Users can interact with the tool using natural language prompts in English, eliminating the need for technical programming skills.
- **Reduces Repetitive Tasks:** Minimizes manual work such as copying and pasting data, allowing users to focus on higher-value activities.

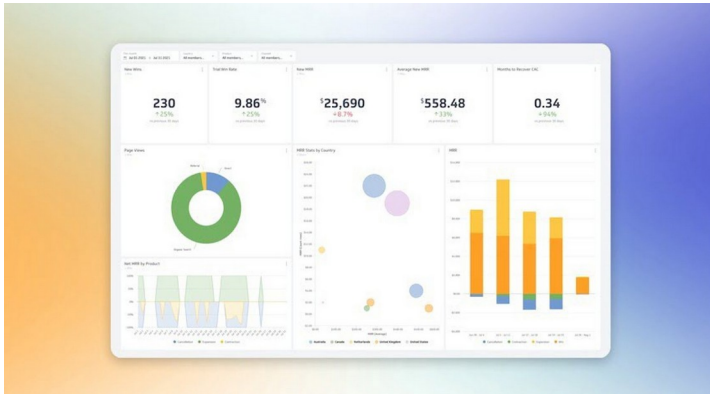
Cons

- **Privacy and Security Risks:** Granting the tool access to your browser may introduce potential privacy and security concerns.
- **Potential for Inaccuracies:** The AI may occasionally generate incorrect information or make assumptions when sufficient information is unavailable.
- **Challenging Fine-Tuning:** Refining specific details can be frustrating, as the AI may not always fully incorporate previous instructions or corrections, requiring multiple iterations to achieve the desired result.

Traditional Dashboard Development vs. AI-Assisted Development

Traditional Dashboards

- Higher development effort
- Manual configuration and maintenance
- Time-consuming design updates
- Changes applied component by component



AI Dashboards

- Faster development cycle
- Prompt-based configuration
- Instant global design updates
- Changes applied across the entire dashboard simultaneously



Key Takeaways from Building Dashboards with AI

Lessons Learned

- **Fast development speed** significantly reduces dashboard creation time.
- **Clean and professional visual layouts** can be generated with minimal effort.
- **Strong coding assistance** helps accelerate development and troubleshooting.

Limitations

- Weak storytelling and report flow may **require manual refinement**.
- May perform tasks **beyond the original request**, requiring careful review.
- Can repeat **previous responses** or solutions, even when alternative approaches are needed.

Key Takeaways from Building Dashboards with AI (cont.)

Unexpected Capabilities

- **Browser Interaction** AI can access web-based resources and perform certain tasks through browser tools.
- **Automatic Dashboard Layout** AI can independently organize charts and visualizations based on the underlying data structure, reducing manual design effort.

Important Considerations

- **Data Sources** Ensure that all information is derived from reliable and verifiable sources.
- **Visual Design** Review chart colors and formatting carefully to ensure they appropriately represent the underlying data and align with reporting standards.

Key Takeaways from Building Dashboards with AI (cont.)

Key Challenges and Limitations

- **Context Window (Token) Limits** Large or complex projects may exceed processing limits, requiring tasks to be split into smaller components and potentially increasing development time.
- **Design Alignment** AI-generated layouts may not always match the desired structure, requiring iterative refinement and clear design specifications.

Areas Requiring Attention

- **Data Sources** Verify that information is obtained from reliable, transparent, and trusted sources.
- **Color Usage** Review visual elements carefully, as AI may select colors that do not appropriately represent the underlying data or intended message.

Recommendation

- **Define Dashboard Scope Explicitly**
 - Detail the exact topics and metrics required for the dashboard.
 - Prevent the AI from drifting outside project boundaries.
- **Validate AI Sources Rigorously**
 - Double-check all citations, data points, and references provided by the AI.
 - Ensure complete alignment and accuracy before updating the skill.md file.
- **Optimize Prompt Efficiency**
 - Think critically and structure requests thoroughly before executing a prompt.
 - Eliminate vague queries to reduce token waste and API overhead.
- **Enforce Strict Design and Color Protocols**
 - Provide explicit hexadecimal codes or predefined color palettes to the AI.
 - Avoid automated color selection to prevent misleading data visualizations.
- **Maintain Human Oversight and Scope Control**
 - Treat AI-generated content as a baseline, not a final solution.
 - Verify all outputs against the specific project scope and technical requirements.

Thank you
